6. In the expansion of $(167)^{131}$ the unit place digit is :–

(A) 7 (B) 9 (C) 1 (D) 3

7. If $|z_1| = 2, |z_2| = 3, |z_3| = 4$ and $|2z_1 + 3z_2 + 4z_3| = 9$, then value of $|8z_2z_3 + 27z_3z_1 + 64z_1z_2|$is equal to :-

(A) 216 (B) 108

(C) 42 (D) 56

8. Let $f(x) = \max\{\sin^{-1}x, \cos^{-1}x\}$ then which of the following is not correct

(A) f(x) is not differentiable at $x = \frac{1}{\sqrt{2}}$

(B) f(x) is continuous everywhere

(C) Range of f(x) is $\left[\frac{\pi}{2}, \pi\right]$

(D) f(x) is many one function

9. The solution of differential equation, $e^{-x}(y + 1)dy + y(\cos^2x - \sin2x)\,dx = 0$ subject to condition $y = 1$ when $x = 0$, is :

(A) $(y + 1) + e^x \cos^2x = 3$

(B) $\ell n(y + 1) + e^x.\cos^2x = 1 + \ell n2$

(C) $y + \ell ny = e^x\cos^2x$

(D) $y + \ell ny + e^x.\cos^2x = 2$

10. If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} = \ln(f(x)) + g(x) + C$ where C is the constant of integration and f(x) is positive, then f(x) + g (x) has the value equal to

(A) $e^x + \sin x + 2x$ (B) $e^x + \sin x$

(C) $e^x - \sin x$ (D) $e^x + \sin x + x$

11. If $\sin^{-1}\sqrt{(x^2 + 2x + 1)} + \sec^{-1}\sqrt{(x^2 + 2x + 1)} = \frac{\pi}{2}$, $x \neq 0$, then the value of $2\sec^{-1}\left(\frac{x}{2}\right) + \sin^{-1}\left(\frac{x}{2}\right)$ is equal to

(A) $-\frac{3\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) $\frac{\pi}{2}$

12. Let a = Minimum $\{x^2 + 2x + 3, x \in R\}$ and $b = \lim_{\theta\to0} \frac{1 - \cos\theta}{\theta^2}$. The value of $\sum_{r=0}^{n} a^r.b^{n-r}$ is:-

(A) $\frac{2^{n+1} - 1}{3.2^n}$ (B) $\frac{2^{n+1} + 1}{3.2^n}$

(C) $\frac{4^{n+1} - 1}{3.2^n}$ (D) None of these

13. Two circles whose radii are equal to 4 and 8 intersect at right angles. The length of their common chord is-

(A) $\frac{16}{\sqrt{5}}$ (B) 8

(C) $4\sqrt{6}$ (D) $\frac{8\sqrt{5}}{5}$

14. A chord is drawn through the focus of the parabola $y^2 = 6x$ such that its distance from the vertex of this parabola is $\frac{\sqrt{5}}{2}$, then its slope can be :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{5}}{2}$

(C) $\frac{2}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{5}}$

15. The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ has its conjugate axis 5 and passes through the point $(2, -1)$. The length of a latus rectum is :

(A) $\frac{3}{4}\sqrt{29}$ (B) $\frac{5}{8}\sqrt{29}$

(C) $\sqrt{29}$ (D) $\frac{5}{4}\sqrt{29}$

16. A line makes angle 45°, 60°, and 60° with the coordinate axis, the projection of line segments on line which joins point $(-1,2,3)$ and $(-1, 4, 0)$ are-

(A) 3/2 (B) 1/3

(C) 1/2 (D) 2/3

17. If $|\vec{c}|^2 = 60$ and $\vec{c} \times (\hat{i} + 2\hat{j} + 5\hat{k}) = \vec{0}$, then a value of $\vec{c}.(-7\hat{i} + 2\hat{j} + 3\hat{k})$ is :

(A) $4\sqrt{2}$ (B) 24

(C) $12\sqrt{2}$ (D) 12

18. Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ and $\beta = \sum_{x \in s} \tan^2\left(\frac{x}{3}\right)$, then $\frac{1}{6}(\beta - 14)^2$ is equal to

(A) 32 (B) 8 (C) 64 (D) 16

19. Let μ be the mean and σ be the standard deviation of the distribution

x_i	0	1	2	3	4	5
f_i	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

where $\sum f_i = 62$. if [x] denotes the greatest integer $\le x$, then $[\mu^2 + \sigma^2]$ is equal

(A) 8 (B) 7 (C) 6 (D) 9

20. Let in a right angled triangle, the smallest angle be θ. If a triangle formed by taking the reciprocal of its sides is also a right angled triangle, then $\sin\theta$ is equal to :

(A) $\frac{\sqrt{5}+1}{4}$ (B) $\frac{\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{2}-1}{2}$ (D) $\frac{\sqrt{5}-1}{4}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : −1 If wrong answer is entered.

---

1. The greatest integer less than or equal to the sum of first 100 terms of the sequence $\frac{1}{3}, \frac{5}{9}, \frac{19}{27}, \frac{65}{81}, \ldots$ is equal to

2. If $|A|_{3\times3} = 9$, then value of $\left|\text{adj}\left(\frac{A}{3}\right)^{-1}\right|$ is -

3. The area enclosed by the curve $xy^2 = 4(2 - x)$ and y-axis is λ square unit, then the value of $8\cos(2014\,\lambda)$ is equal to

4. Number of values of x satisfying the equation $\int_{-1}^{x}\left(8t^2 + \frac{28}{3}t + 4\right)dt = \frac{\left(\frac{3}{2}\right)x + 1}{\log_{(x+1)}\sqrt{x+1}}$, is :

5. Let $f(n) = \left[\frac{1}{3} + \frac{3n}{100}\right]n$, where [n] denotes the greatest integer less than or equal to n. Then $\sum_{n=1}^{56} f(n)$ is equal to :-

6. If $\lim_{x\to0}\frac{ax - (e^{4x} - 1)}{ax(e^{4x} - 1)}$ exists and is equal to b, then the value of $a - 2b$ is _____.

7. If A and B are two sets such that $A = \{1, 2, x\}$, $B = \{3, 4, y\}$ and $\{1, 3\} \times \{2, 4\} \subseteq A \times B$ then the value of $x + y$ is :

8. The integral value of a for which the point $(-2, a)$ lies in the interior of the triangle formed by the lines $y = x$, $y = -x$ and $2x + 3y = 6$.

9. If the lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then 2k is equal to :

10. $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Main)
SCORE - II
13-02-2024

JEE(Main) : ENTHUSIAST COURSE (SCORE-II)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	A	D	C	B	D	A	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	D	C	C	D	B	C	C	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	20	40	90	2	45	660	44	83	3	200

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	C	B	B	A	D	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	B	B	B	A	D	C	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	140	1	350	5	11	2	4	3	5	32

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	C	A	D	A	C	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	A	B	D	C	C	A	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	98	9	8	1	1399	5	5	3	9	3

HINT – SHEET

PART-1 : PHYSICS

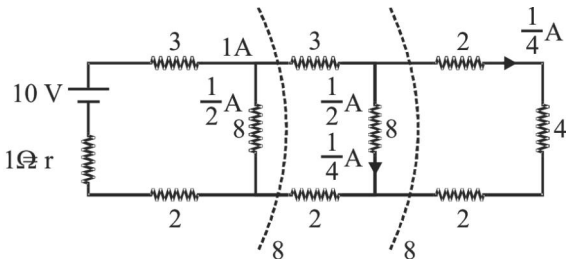
SECTION-I

1. Ans (D)

Equivalent resistance R_{eq} = 10 Ω so current passing through battery and 3 Ω resistance is

$i = \frac{10}{10}$ = A

and current passing through 4 Ω is 0.25 A



2. Ans (D)

$V_C = V_{6\Omega} = i \times 6 = \left(\frac{12}{2+6}\right) \times 6 = 9V$

$q = CV$

$= 2 \times 9$

$= 18\ \mu C$

3. Ans (A)

$\tau = MB \sin\theta \Rightarrow \tau = (mL)\, B \sin\theta$

$\Rightarrow \quad 25 \times 10^{-6} = (m \times 5 \times 10^{-2}) \times 5 \times 10^{-2} \times \sin 30$

$\Rightarrow \quad m = 2 \times 10^{-2}$ A – m

4. Ans (D)

No flux change is taking place because magnetic field exists everywhere and is constant in time and space.

5. Ans (C)

$P = V_{r.m.s.} \times i_{r.m.s.} \times \cos\phi$

$= \frac{100}{\sqrt{2}} \times \frac{100 \times 10^{-3}}{\sqrt{2}} \times \cos\frac{\pi}{3}$

$= \frac{10^4 \times 10^{-3}}{2} \times \frac{1}{2} = \frac{10}{4} = 2.5$ watt

6. Ans (B)

$\vec{E} \times \vec{B} = \hat{K}$

$\vec{E}.\vec{B} = 0$

7. Ans (D)

$V = \frac{A\sqrt{x}}{x + B}$

$B = L \quad A = \frac{ML^2T^{-2}L}{\sqrt{L}}$

$A = ML^{5/2}T^{-2}$

$AB = ML^{7/2}T^{-2}$

8. Ans (A)

$x = \mu\cos\theta t \quad y = \mu\sin\theta t - \frac{1}{2}gt^2$

$x = 36t \qquad y = \frac{96}{2}t - \frac{1}{2}(9.8)t^2$

$\mu\cos\theta = 36$

$\mu\sin\theta = \frac{96}{2}$

$\therefore \quad \tan\theta = \frac{96/2}{36} = \frac{4}{3}$

$\therefore \quad \sin\theta = \frac{4}{5}$

9. Ans (B)

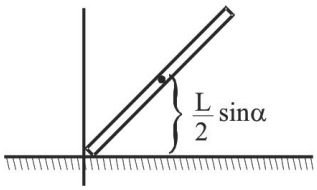
$S = \frac{1}{2}a.4^2$

$\frac{S}{4} = \frac{1}{L}at^2$

$4 = \frac{4^2}{t^2}$

$t = 2$ sec.

10. Ans (A)



$Mg\frac{L}{2}\sin\alpha = \frac{1}{2}I\omega^2$

$Mg\frac{L}{2}\sin\alpha = \frac{1}{2}\frac{ML^2}{3}\omega^2$

$\therefore w = \sqrt{\frac{3g\sin\alpha}{L}}.$

11. Ans (C)

$V_0 = \sqrt{\frac{GM}{r}} \Rightarrow V_0 \propto \frac{1}{\sqrt{r}}$

$\frac{V_1}{V_2} = \sqrt{\frac{r_2}{r_1}}$

$\frac{V}{V_2} = \sqrt{\frac{4R}{R}}, V_2 = \frac{V}{2}$

12. Ans (C)

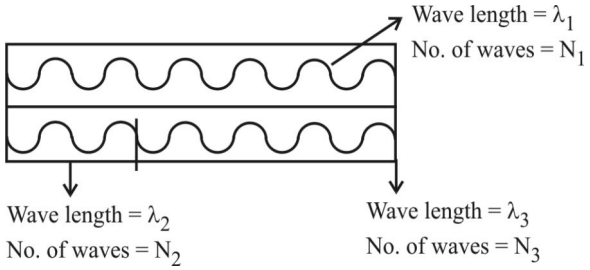
Total force at height 3L/4 from its lower end

= Weight suspended + Weight of 3/4 of the wire

$= W_1 + (3W/4)$

Hence stress $= \frac{W_1 + (3W/4)}{S}$

13. Ans (D)



$\lambda_1 = \frac{\lambda_0}{\mu_1}$

Also, $\lambda_1 \times N_1 = 3x$

$\therefore\ N_1 = \frac{3x}{\lambda_0}\mu_1$

Similarity $N_2 = \frac{x}{\lambda_0}\mu_2$

$N_3 = \frac{2x}{\lambda_0}\mu_3$

$N_1 = N_2 + N_3$

$\frac{3x \times 1.5}{\lambda_0} = \frac{x}{\lambda_0} \times \mu_2 + \frac{2x}{\lambda_0} \times 1.4$

$4.5 = \mu_2 + 2.8$

$\mu_2 = 4.5 - 2.8 = 1.7$

14. Ans (C)

$x = \frac{b}{2} = (2n-1)\frac{\lambda D}{2d}$

$\lambda = \frac{b^2}{(2n-1)\,d}$

15. Ans (C)

From photoelectric equation

$V_1 = \frac{hc}{e\lambda_1} - \frac{\phi}{e}$

$V_2 = \frac{hc}{e\lambda_2} - \frac{\phi}{e}$

$V_3 = \frac{hc}{e\lambda_3} - \frac{\phi}{e}$

$\because$ V_1, V_2 & V_3 are in A.P.

$\Rightarrow\ 2V_2 = V_1 + V_3$

$\Rightarrow \frac{2hc}{e\lambda_2} - \frac{2\phi}{e} = \frac{hc}{e\lambda_1} = \frac{\phi}{e} + \frac{hc}{e\lambda_3} - \frac{\phi}{e}$

$\Rightarrow \frac{2}{\lambda_2} = \frac{1}{\lambda_1} + \frac{1}{\lambda_3}$

$\Rightarrow$ λ_1, λ_2 and λ_3 are in HP.

16. Ans (D)

Given E_1 the binding energy (BE) per nucleon of deutron (2-nucleons)

So, the BE of deutron = $2E_1$

Also, given BE per nucleon of an a-particle (4-nucleons) = E_2

So, the BE of α-particle = $4E_2$

In the given nuclear reaction,

$${}_1H^2 + {}_1H^2 \rightarrow {}_2H^4 + E$$

Using energy conservation law,

$2E_1 + 2E_1 = 4E_2 + E$

or $E = 4E_1 - 4E_2 = 4(E_1 - E_2)$

$\therefore$ Energy E released = $-E = 4(E_2 - E_1)$

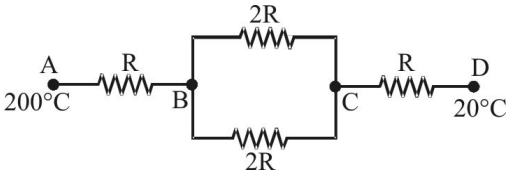
18. Ans (C)

Output equation $y = \overline{\overline{A} + \overline{B}} = \overline{A.B}$

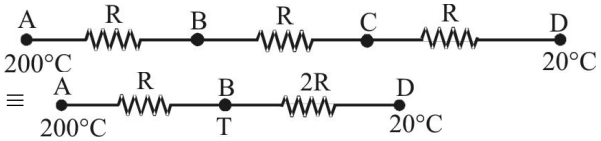
19. Ans (C)

Heat absorbed in (3)

20. Ans (C)



$\equiv$



$\equiv$

$\frac{200 - T}{R} = \frac{T - 20}{2R}$

$400 - 2T = T - 20$

$T = 140°C$

PART-1 : PHYSICS

SECTION-II

1. Ans (20)

$(E_x) = -\frac{\partial v}{\partial x}\hat{i} = -6\hat{i}$

(0,0,0)

$(E_y) = -\frac{\partial v}{\partial y}\hat{j} = -18\hat{j}$

(0,0,0)

$(E_z) = -\frac{\partial v}{\partial z}\hat{k} = 0\hat{k}$

$|E| = \sqrt{6^2 - 18^2} = 10$

$F = qE = 2$

2. Ans (40)

In balancing

$\frac{R}{S} = \frac{25}{75}$

New resistance $R' = \frac{\rho\ell}{A}$

$= \frac{\rho \times \frac{\ell}{2}}{\frac{A}{4}} = \frac{\rho\ell}{2} \times 4A$

$R' = 2R$

$\frac{2R}{S} = \frac{\ell'}{100 - \ell'}$

$2 \times \frac{1}{3} = \frac{\ell'}{100 - \ell'} = 3\ell' = 200 - 2\ell'$

$5\ell' = 200$

$\ell' = 40$

$\therefore$ Correct answer 40

3. Ans (90)

$\vec{F} = \frac{d\vec{p}}{dt} = -2\sin t\hat{i} + 2\cos t\hat{j}$

$\vec{P} = 2\cos t\hat{i} + 2\sin t\hat{j}$

$\vec{F} \cdot \vec{P} = -4\cos t\sin t + 4\cos t\sin t = 0$

$\therefore \quad \theta = 90°$

4. Ans (2)

$U = \frac{16}{x^{12}} - \frac{1}{2x^6}$

$F = -\frac{dU}{dx} = 0, \quad x = 2$

6. Ans (660)

$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$

$1.6\pi = \frac{2\pi}{\lambda} \times \frac{40}{100} \Rightarrow \lambda = 1/2 \text{ m}$

$\therefore f = \frac{v}{\lambda} = \frac{330}{1/2} = 660 \text{ Hz}$

7. Ans (44)

$\frac{6}{0.125} = R + \frac{1}{\left(\frac{1}{8} + \frac{1}{16} + \frac{1}{16}\right)}$

8. Ans (83)

${}_{84}X^{202} \xrightarrow{\alpha\text{-decay}} {}_{82}Y^{198} + {}_2He^4$

${}_{82}Y^{198} \xrightarrow{\beta\text{-decay}} {}_{83}Z^{198} + {}_{-1}\beta^0$

9. Ans (3)

$\gamma_{mix} = \frac{n_1Cp_1 + n_2Cp_2}{n_1Cv_1 + n_2Cv_2}$

$\frac{11}{7} = \frac{n_1\left(\frac{5}{2}R\right) + n_2\left(\frac{7}{2}R\right)}{n_1\left(\frac{3}{2}R\right) + n_2\left(\frac{5}{2}R\right)}$

$\frac{n_1}{n_2} = 3$

10. Ans (200)

$\frac{T_2}{T_1} = \frac{\lambda m_1}{\lambda m_2} = \frac{1.75}{14.35} = T_2 = \frac{1.75}{14.35} \times 1640 = 200K$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B)

For exothermic process as T↑ yield ↓

so $T_1 > T_2 > T_3$

2. Ans (A)

$k_1 = \frac{2.303}{t} \times \log\left(\frac{100}{100 - 90}\right)$

$= \frac{2.303}{t} \quad (\log 10 = 1)$

$k_2 = \frac{2.303}{2t} \times \log\left(\frac{100}{100 - 99}\right)$

$= \frac{2.303}{t} (\log 100 = 2)$

$\therefore \frac{k_1}{k_2} = 1$

4. Ans (C)

$pH = \frac{1}{2}(pKa - \log c) = \frac{1}{2}(4 - (-1)) = 2.5$

$pH = pKa + \log \frac{[A^-]}{[HA]} = 4 + 0 = 4$

$pH = 7 + \frac{1}{2}(pKa + \log c) = 7 + \frac{1}{2}(4 + \log \frac{2}{30})$

$= 8.41$

5. Ans (B)

$$\frac{\Delta T_b}{\Delta T_f} = \frac{K_b \times m}{K_f \times m}$$

$$\frac{y}{\Delta T_f} = \frac{x \times m}{z \times m}$$

$$\Delta T_f = \frac{y \cdot z}{x}$$

6. Ans (B)

For max, rise in temp. ; max. neutralization of H^+ and OH^- required.

If we take equal volume, all H^+ (5 m-mole) will react with all OH^- (5 m-mole)

7. Ans (A)

$\Delta G^\circ = -RT\ell nK$

$$= \frac{-8.314 \times 298 \times \ell n}{1000} \frac{(10^{-2})^2}{10^{-2}(10^{-5})^2}$$

$$= \frac{-8.314 \times 298 \times \ell n}{1000}(10^8)$$

$= -8.314 \times 298 \times 8 \times 2.3/1000$

$= -45.65$ kJ

9. Ans (D)

$Si - O$

$SiH_3 - O - SiH_3$

$(CH_3)_3Si-O^{\Theta}$

They all have $p\pi - d\pi$ back bonding.

11. Ans (B)

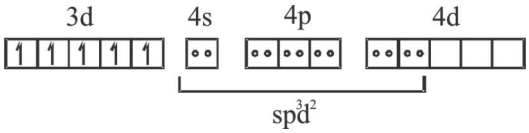
$\left[\overset{+3}{Cr}(NH_3)_2Cl_2(C_2O_4)\right]^-$ $[CrCl_6]^{-3}$

Co-ordination no. = 6 Charge = −3

12. Ans (A)

$[FeF_6]^{3-}$

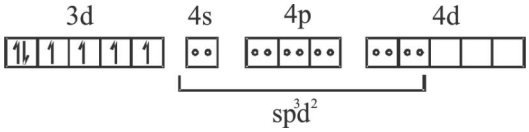
$Fe^{+3} = [Ar]\ 4s^0 3d^5$



Number of unpaired $e^- = 5$

$[CoF_6]^{3-}$

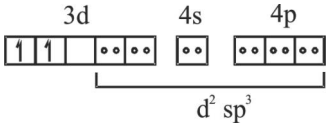
$Co^{+3} = [Ar]4s^\circ 3d^6$



Number of unpaired $e^- = 4$

$[V(H_2O)_6]^{3+}$

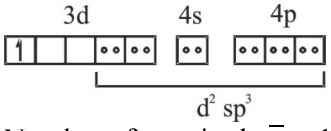
$V^{+3} = [Ar]\ 4s^\circ 3d^2$



Number of unpaired e– = 2

$[Ti(H_2O)_6]^{3+}$

$Ti^{+3} = [Ar]4s^\circ 3d^1$



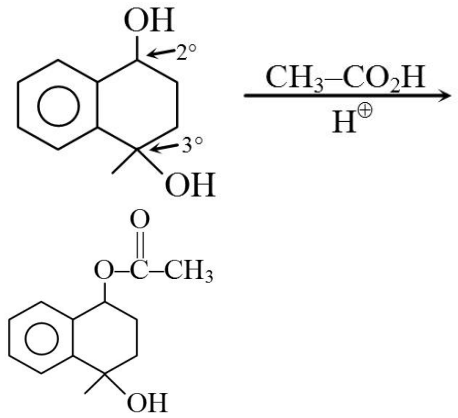
Number of unpaired $e^- = 1$

14. Ans (B)

$Y^{3+} < La^{3+}$

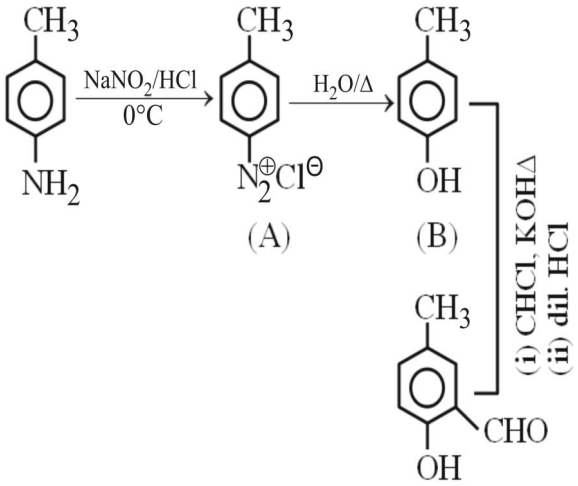
$Lu^{3+} < Eu^{3+} < La^{3+}$ } (Order of radii)

15. **Ans (B)**

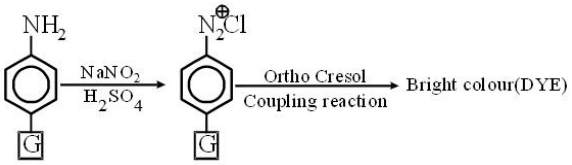


Rate of esterification $\propto \frac{1}{\text{steric hindrance}}$

16. **Ans (A)**

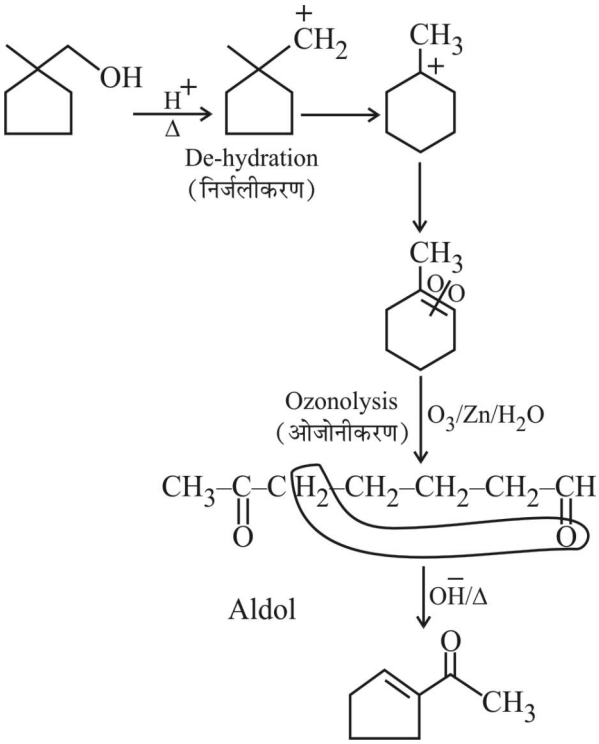


17. **Ans (D)**

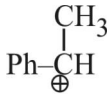


G must be electron releasing group like CH_3

18. **Ans (C)**



19. **Ans (B)**

According to stable carbocation of  reaction follows S_N1 & due to chiral centre recemic mixture is form.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (140)**

$$k = \left(\frac{k_1 . k_2}{k_3}\right)^{2/3}$$

$$= \left[\frac{\left(A_1 \cdot e^{-\frac{Ea_1}{RT}}\right) . \left(A_2 \cdot e^{-\frac{Ea_2}{RT}}\right)}{\left(A_3 \cdot e^{-\frac{Ea_3}{RT}}\right)}\right]^{2/3}$$

$$= \left[A' . e^{-\frac{Ea_1}{RT}-\frac{Ea_2}{RT}+\frac{Ea_3}{RT}}\right]^{2/3}$$

$$= (A')^{2/3} . \ e^{\left[\frac{-Ea_1+Ea_2-Ea_3}{RT}\right]^{2/3}}$$

On comparison with Arhenius Eq :

$$E_a = \frac{2}{3}[Ea_1 + Ea_2 - Ea_3]$$

$$\Rightarrow Ea = \frac{2}{3}[180 + 80 - 50] = 140 \text{ kJ}$$

3. **Ans (350)**

$$\left(200 = \frac{P_i}{4} + \frac{P_P \times 3}{4}\right) \quad ....(1)$$

$$\left(300 = \frac{3P_i}{4} + \frac{P_P}{4}\right) \times 3 \quad ....(2)$$

$$700 = \frac{9}{4}P_i - \frac{P_i}{4} = 2P_i$$

$$P_i = \frac{700}{2} = 350 \text{ mm of Hg.}$$

4. **Ans (5)**

The chemical reaction occurring here is

$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$

Now, moles of NaCl given = $\frac{4.77}{58.5} = 0.0815$

& moles of $AgNO_3 = \frac{5.77}{170} = 0.0339$

Here, $AgNO_3$ is the limiting reagent and moles of AgCl formed = Moles of $AgNO_3$

Mass of AgCl = 0.0339 × mol. wt. of AgCl

= 0.0339 × 143.5 = 4.87 g

5. **Ans (11)**

$X = KMnO_4$

$Y = MnO_2$

6. **Ans (2)**

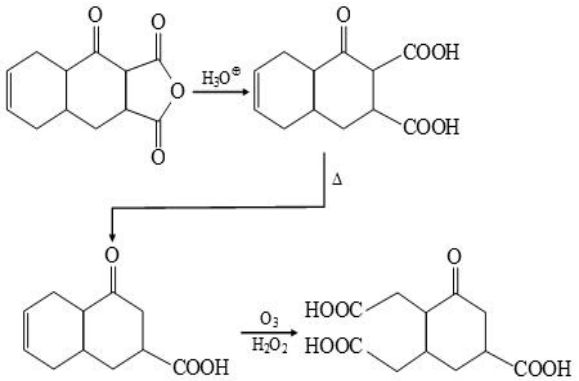
$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_4 + 3CH_4$

$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$

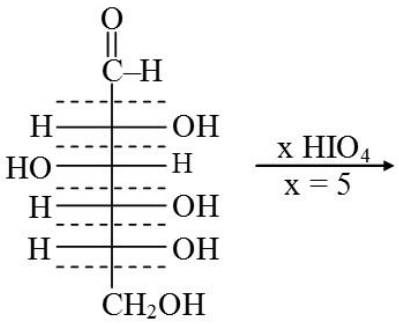
7. **Ans (4)**

(b), (c), (d) and (e) are correct.

8. **Ans (3)**



9. **Ans (5)**



Each C–C bond cleavage require 1 HIO_4

10. **Ans (32)**

$\boxed{32\alpha - H}$

(1) 7α–H

(2) 8α–H

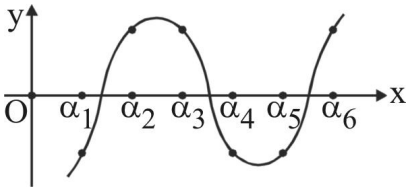
(3) 7α–H

(4) 10α–H

PART-3 : MATHEMATICS

SECTION-I

2. **Ans (D)**



No real root in $(-\infty, \alpha_1)$

3. **Ans (B)**

$1800 = 2^3 . 3^2 . 5^2$

Total no. of divisor which are multiple of 10 is

= (2 + 1) (2 + 1) (1 + 1) = 18

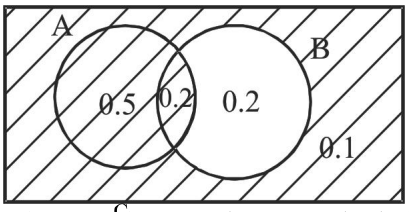
but in proper divisor 1800 must be excluded so proper divisors = 18 – 1 = 17

4. **Ans (C)**

Required ways

$${}^9C_6 \times \frac{\lfloor 6}{\lfloor 3 \lfloor 3} - {}^{3\times 6}C_6 \times \frac{\lfloor 6}{\lfloor 3 \lfloor 3} = 1620$$

5. Ans (A)



$P(A \cup B^C)$ = Total case = Shaded region

so $P\left[\frac{B}{A \cup B^C}\right] = \frac{0.2}{0.8} = \frac{1}{4}$

6. Ans (D)

$(167)^{131} = (170 - 3)^{131} \Rightarrow (3^{131})$

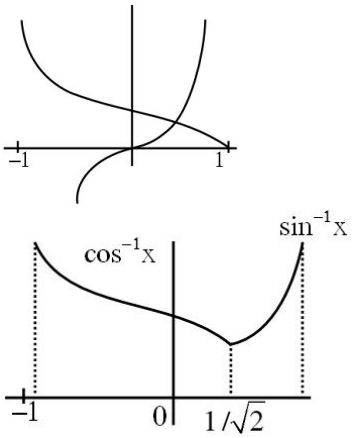
$\Rightarrow -(3^2)^{165}$

$\Rightarrow -3(3^2)^{165}$

$\Rightarrow -3(10 - 1)^{165}$

$\Rightarrow -3\,(-1)^{165} = 3$

8. Ans (C)



9. Ans (D)

$\left(\frac{y+1}{y}\right) dy + e^x\left(\cos^2 x - \sin 2x\right) dx = 0$

$\therefore y + \ell ny + e^x.\cos^2 x = C$

As, $y(0) = 1 \Rightarrow C = 2$

10. Ans (B)

$$\int \frac{(e^x + \cos x + 1) - (e^x + \sin x + x)}{e^x + \sin x + x}\, dx$$

$$= \int \frac{e^x + \cos x + 1)}{e^x + \sin x + x}\, dx - \int dx$$

$= \ln(e^x + \sin x + x) - x + C$

11. Ans (B)

$\because$

$\sin^{-1}\sqrt{(x^2 + 2x + 1)} + \sec^{-1}\sqrt{(x^2 + 2x + 1)} = \frac{\pi}{2}$

or $\sin^{-1}|x + 1| + \cos^{-1}\left(\frac{1}{|x+1|}\right) = \frac{\pi}{2}$

or $|x + 1| = \frac{1}{|x+1|}$ $(x \neq -1)$

$\Rightarrow |x + 1|^2 = 1$

$\Rightarrow |x + 1| = 1$

$\Rightarrow x + 1 = \pm 1$

or $x = 0, -2$

But given $x \neq 0$

$\therefore x = -2$

Then, $2\sec^{-1}\left(\frac{x}{2}\right) + \sin^{-1}\left(\frac{x}{2}\right)$

$= 2\sec^{-1}(-1) + \sin^{-1}(-1)$

$= 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$

12. Ans (C)

$a = 2, \quad b = \frac{1}{2}$

$$\sum_{r=0}^{n} 2^r \left(\frac{1}{2}\right)^{n-r} = \frac{4^{n+1} - 1}{3.2^n}$$

14. Ans (B)

Eq of chord

$y - 0 = m\,(x - 3/2)$

$$\left|\frac{-\frac{3}{2}m}{\sqrt{m^2 + 1}}\right| = \frac{\sqrt{5}}{2} \Rightarrow m = \frac{\sqrt{5}}{2}$$

15. Ans (D)

$b = \frac{5}{2}$, $(2, -1)$ lies on $\frac{x^2}{a^2} - \frac{4y^2}{25} = 1$

$\therefore \frac{4}{a^2} = \frac{29}{25} \Rightarrow a = \frac{10}{\sqrt{29}}$

Latus rectum $\frac{2b^2}{a} = 2\left(\frac{25}{4}\right).\frac{\sqrt{29}}{10} = \frac{5}{4}\sqrt{29}$

16. Ans (C)

$\alpha = 45°,\ \beta = \gamma = 60°$

$\ell = \frac{1}{\sqrt{2}}, m = n = \frac{1}{2}$: $(x_1, y_1, z_1) \equiv (-1, 2, 3)$

$(x_2, y_2, z_2) \equiv (-1, 4, 0)$

Use $|\ell.(x_2 - x_1) + m.(y_2 - y_1) + n.(z_2 - z_1)|$

$\left|0 + \frac{1}{2}(4-2) + \frac{1}{2}(0-3)\right|$

$= \left|1 - \frac{3}{2}\right| = \frac{1}{2}$

17. Ans (C)

$\bar{C} \times (\hat{i} + 2\hat{j} + 5\hat{k}) = 0$

$\vec{C} = \lambda(\hat{i} + 2\hat{j} + 5\hat{k})$

$|C| = \lambda\sqrt{30} \Rightarrow \lambda^2 (30) = |c|^2 = 60$

$\lambda = \pm\sqrt{2}$

$\Rightarrow \bar{C}.(-7\hat{i} + 2\hat{j} + 3\hat{k})$

$\Rightarrow \lambda (i + 2j + 5k) . (-7i + 2j + 3k)$

$\Rightarrow \lambda (-7 + 4 + 15) = 12\lambda$

$= 12\sqrt{2}$ or $-12\sqrt{2}$

18. Ans (A)

Let $9^{\tan^2 x} = P$

$\Rightarrow \frac{9}{P} + P = 10$

$P^2 - 10P + 9 = 0$

$(P - 9)(P - 1) = 0$

$P = 1, 9$

$\Rightarrow 9^{\tan^2 x} = 1\ ,\ 9^{\tan^2 x} = 9$

$\tan^2 x = 0,\ \tan^2 x = 1$

$x = 0,\ \pm\frac{\pi}{4} \therefore x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\beta = \tan^2(0) + \tan^2\left(+\frac{\pi}{12}\right) + \tan^2\left(-\frac{\pi}{12}\right)$

$= 0 + 2(\tan 15°)^2$

$2(2 - \sqrt{3})^2$

$2(7 - 4\sqrt{3})$

Than $\frac{1}{6}(14 - 8\sqrt{3} - 14)^2 = 32$

19. Ans (A)

$\sum f_i = 62$

$\Rightarrow 3k^2 + 16k - 12k - 64 = 0$

$\Rightarrow k = \text{or} -\frac{16}{3}$ (rejected)

$\mu = \frac{\sum f_i x_i}{\sum f_i}$

$\Rightarrow \mu = \frac{8 + 2(15) + 3(15) + 4(17) + 5}{62} = \frac{156}{62}$

$\sigma^2 = \sum f_i x_i^2 - \left(\sum f_i x_i\right)^2$

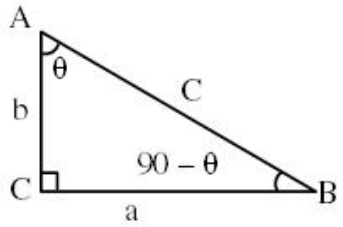
$= \frac{8 \times 1^2 + 15 \times 13 + 17 \times 16 + 25}{62} - \left(\frac{156}{62}\right)^2$

$\sigma^2 = \frac{500}{62} - \left(\frac{156}{62}\right)^2$

$\Rightarrow \sigma^2 + \mu^2 = \frac{500}{62}$

$\left[\sigma^2 + \mu^2\right] = 8$

20. Ans (B)



$<A = \theta$

$<B = 90 - \theta$

a = smallest side

$c^2 = a^2 + b^2$

$\frac{1}{a^2} = \frac{1}{b^2} + \frac{1}{c^2}$

$\frac{b^2c^2}{a^2} = b^2 + c^2$

Use $a = 2R\sin A = 2R\sin\theta$

$b = 2R\sin B = 2R\sin(90 - \theta) = 2R\cos\theta$

$c = 2R\sin C = 2\sin 90° = 2R$

$\frac{4R^2\cos^2\theta}{4R^2\sin^2\theta} = 4R^2\cos^2\theta + 4R^2$

$\cos^2\theta = \sin^2\theta\cos^2\theta + \sin^2\theta$

$1 - \sin^2\theta = \sin^2\theta(1 - \sin^2\theta) + \sin^2\theta$

$\sin^2\theta = \frac{3 - \sqrt{5}}{2}$

$\Rightarrow \sin\theta = \frac{\sqrt{5} - 1}{2}$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (98)

$\frac{1}{3}+\frac{5}{9}+\frac{19}{27}+\frac{65}{81}+\cdots$ 100 terms

$=\left(1-\frac{2}{3}\right)+\left(1-\frac{4}{9}\right)+\left(1-\frac{8}{27}\right)+\left(1-\frac{16}{81}\right)\ldots.100 \text{ terms}$

$=100-\left[\frac{2}{3}+\left(\frac{2}{3}\right)^2+\cdots.\right]$

$=100-\frac{\frac{2}{3}\left(1-\left(\frac{2}{3}\right)^{100}\right)}{1-\frac{2}{3}}$

$=100-2\left(1-\left(\frac{2}{3}\right)^{100}\right)$

$S=98+2\left(\frac{2}{3}\right)^{100} \Rightarrow [S]=98$

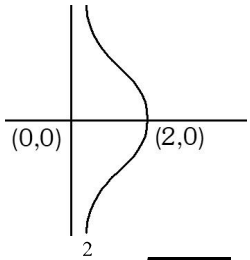
2. Ans (9)

Let $\left(\frac{A}{3}\right)=B$

$|\text{adj}B^{-1}| = |B^{-1}|^2 = \frac{1}{|B|^2} = \frac{1}{\left|\frac{A}{3}\right|^2}$

$=\frac{3^6}{|A|^2}=\frac{3^6}{9^2}=3^2=9$

3. Ans (8)



$\lambda=2\int_0^2 2\sqrt{\frac{2-x}{x}}\,dx$

Put $x=2\sin^2\theta$

$=16\int_0^{\pi/2}\cos^2\theta\,d\theta$

$=16\times\frac{1}{2}\times\frac{\pi}{2}=4\pi$

Then, $\cos(2014\times 4\pi)=1$

$\therefore\ 8\times 1=8$

4. Ans (1)

Integrating LHS a cubic in x of the form $x(ax^2+bx+c)=0$

giving x=0, −3/2,−1/4. Only x = −1/4 satisfies]

5. Ans (1399)

$\sum_{n=1}^{56} f(x)=\left[\frac{1}{3}+\frac{3\times 1}{100}\right]\times 1+\ldots.+\left[\frac{1}{3}+\frac{3\times 22}{100}\right]\times 22$

$+\left[\frac{1}{3}+\frac{3\times 23}{100}\right]\times 23 \ldots\ldots\ldots+\ldots\ldots\ldots$

$\left[\frac{1}{3}+\frac{3\times 55}{100}\right]\times 55+\left[\frac{1}{3}+\frac{3\times 56}{100}\right]\times 56$

$=0+\ldots\ldots\ldots+0+23+24+\ldots\ldots.+55+2\times 56$

$=\frac{55(56)}{2}-\frac{22(23)}{2}+112$

$=11\,(5\times 28-23)+112$

$=11\times 117+112$

$=1287+112$

$=1399$

6. Ans (5)

$\lim_{x\to 0}\frac{ax-\left(e^{4x}-1\right)}{ax\left(e^{4x}-1\right)}\ \left(\frac{0}{0}\right)$

$=\lim_{x\to 0}\frac{ax-\left(e^{4x}-1\right)}{ax\,.\,4x}$ $\boxed{\text{Use } \lim_{x\to 0}\frac{e^{4x}-1}{4x}=1}$

Apply L'Hospital Rule

$=\lim_{x\to 0}\frac{a-4e^{4x}}{8ax}\ \left(\frac{a-4}{0}\text{ form}\right)$

limit exists only when $a-4=0 \Rightarrow a=4$

$=\lim_{x\to 0}\frac{4-4e^{4x}}{32x}$

$=\lim_{x\to 0}\frac{1-e^{4x}}{8x}\ \left(\frac{0}{0}\right)$

$=\lim_{x\to 0}\frac{-e^{4x}.4}{8}=-\frac{1}{2} \Rightarrow b=-\frac{1}{2}$

$a-2b=4-2\left(-\frac{1}{2}\right)$

$=5$

7. Ans (5)

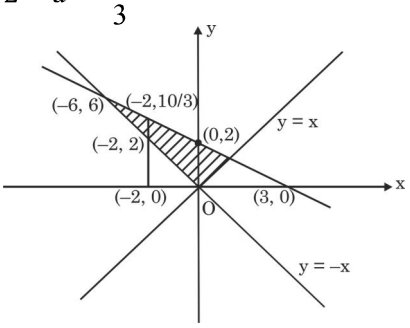
$\{1,3\}\times\{2,4\}\subseteq A\times B$

$\Rightarrow \{(1,2),(1,4),(3,2),(3,4)\}\subseteq\{(1,3),(1,4),(1,y),(2,3),(2,4),(2,y),(x,3),(x,4),(x,y)\}$

$\Rightarrow y=2,\ x=3$

8. Ans (3)

$2 < a < \frac{10}{3}$



9. Ans (9)

$$\begin{vmatrix} 3-1 & K+1 & 0-1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 2 & K+1 & -1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$\Rightarrow 2K - 9 = 0$

$\Rightarrow K = \frac{9}{2}$

10. Ans (3)

$P = 96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$

$2P \times \sin\frac{\pi}{33} = 96 \times 2\sin\frac{\pi}{33}\cos$

$\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$

$2P \times \sin\frac{\pi}{33} = 6 \times \sin\frac{32\pi}{33} = 6\sin\frac{\pi}{33}$

$P = 3$